AN ELEMENTARY DISTINGUISHED SQUARE IS A PUSHOUT SQUARE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1723-1735 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 1737-1761, the complete author proof (23 body lines).

Editorial note. Navigation stubs: 0 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 32 of queue_spaces-pushouts.json (pb=907, stmt 1723-1735, proof 1737-1761).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Lemma 0.1. *Let S be a scheme. Let $(U \subset W, f : V \to W)$ be an elementary distinguished square. Then*

$$\begin{array}{ccc} U \times_W V & \longrightarrow & V \\ \downarrow & & \downarrow f \\ U & \longrightarrow & W \end{array}$$

is a pushout in the category of algebraic spaces over S.

Proof. Observe that $U \amalg V \to W$ is a surjective étale morphism. The fibre product

$$(U \amalg V) \times_W (U \amalg V)$$

is the disjoint union of four pieces, namely $U = U \times_W U$, $U \times_W V$, $V \times_W U$, and $V \times_W V$. There is a surjective étale morphism

$$V \amalg (U \times_W V) \times_U (U \times_W V) \longrightarrow V \times_W V$$

because f induces an isomorphism over $W \setminus U$ (part of the definition of being an elementary distinguished square). Let B be an algebraic space over S and let $g : V \to B$ and $h : U \to B$ be morphisms over S which agree after restricting to $U \times_W V$. Then the description of $(U \amalg V) \times_W (U \amalg V)$ given above shows that $h \amalg g : U \amalg V \to B$ equalizes the two projections. Since B is a sheaf for the étale topology we obtain a unique factorization of $h \amalg g$ through W as desired. □

Date: stacks-project:spaces-pushouts.tex:local-snapshot-20261012; GNU FDL 1.2 or later.